

# Differential Splicing with Subisoform Covariance

## Overview

### Questions and unit of analysis

The analysis asks two distinct questions:

1. Does local splice-path usage differ by experimental condition within a cell type?
2. Does local splice-path usage differ across cell types within the same mice?

The primary analysis aggregates cells into *mouse-by-condition-by-cell-type pseudobulks*. An evaluated alternative retains each cell's EC counts and isoform baseline in the local likelihood but estimates one shared path effect for the pseudobulk. Both approaches keep biological replication at the mouse level; cells are never treated as independent condition replicates.

A splice block is a local region between constitutive exonic anchors. Transcripts that take the same path through that region are collapsed to one subisoform. This targets local splicing choices, which can be identifiable even when the corresponding full-length isoforms are not.

For condition effects, each hypothesis is indexed by a splice block and a cell type. It is an omnibus test of

$$H_0 : \text{mean path usage is equal across the included conditions.}$$

Only conditions represented by at least three mouse pseudobulks are included. With two conditions and two paths, this is a precision-weighted differential-PSI test. With more conditions or paths, all condition coefficients and path log-ratios are tested jointly.

For cell-type effects, each hypothesis is indexed only by a nonredundant splice-path partition. It is an omnibus test of equal path usage across the included cell types, after absorbing mouse and condition into a mouse-specific fixed effect. Cell types must be represented in at least three mice, and the primary FDR family requires at least 30 pseudobulk observations per partition. These two hypothesis families are reported separately.

## Workflow

The analysis proceeds in five steps:

1. Construct local splice blocks from the isoform annotation and collapse isoforms to paths within each block.
2. Aggregate paired oligo(dT) and random-hexamer EC counts, and hierarchical isoform estimates, within each mouse, condition, and cell type.
3. Refine path usage in each pseudobulk with the exact paired-primer EC likelihood. Optionally, retain separate cell likelihood terms while sharing one pseudobulk path perturbation.

4. Estimate path-logratio covariance from EC ambiguity and UMI sampling. Estimates with weak information or boundary path usage are retained in output but excluded from Gaussian testing.
5. Fit mouse-replicate GLS condition models and compositional cell-type models. Test condition within cell type and cell type within mouse as separate hypothesis families.

Two inferential covariance calculations are reported. The *conditional* calculation fixes the fitted isoform mixture within each path. The *profile* calculation allows all isoform directions to vary as nuisance parameters. They are sensitivity analyses for the same block-by-cell-type hypothesis, not independent biological hypotheses.

## Notation and Dimensions

An *equivalence class* (EC) is a set of isoforms compatible with one UMI. An EC count is therefore observed, whereas its allocation among compatible isoforms is latent. A *splice path* is an ordered local exon chain between two constitutive anchors. A *subisoform* is the set of full-length isoforms taking the same path through one block. PSI means percent spliced in; for two paths it is one component of the path-composition vector. ILR means isometric log-ratio, an orthonormal coordinate system for a composition.

Let  $g = g(b)$  denote the gene containing block  $b$ , and let  $\mathcal{I}_g$  be the set of isoforms retained by the paired-primer model and assigned to gene  $g$  by the isoform-to-gene table. Define  $T_g = |\mathcal{I}_g| \in \mathbb{N}$ . This is a gene-level set, not the set of isoforms whose exons overlap the block, and it is therefore shared by every block of the gene. Let  $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$  be the retained, annotated spliced isoforms represented in the block's GTF-derived path map. Isoforms in  $\mathcal{I}_g \setminus \mathcal{I}_b^{\text{sp}}$ , including modeled unspliced precursors, remain nuisance isoforms in the EC likelihood.

Let  $n_b, S_b \in \mathbb{N}$  be the numbers of qualifying pseudobulks and represented spliced paths for block  $b$ , with  $S_b \geq 2$ . Then  $d_b = S_b - 1 \in \mathbb{N}$  is its number of independent log-ratios. There are  $P = 2$  primer protocols. Primer  $p$  has  $E_{bp} \in \mathbb{N}$  ECs for gene  $g(b)$ . Pseudobulk  $i$  contains  $H_{bi} \in \mathbb{N}$  cells. The principal symbols are:

| symbol                     | dimension or domain                | meaning                           |
|----------------------------|------------------------------------|-----------------------------------|
| $b$                        | $\{1, \dots, \mathcal{B}\}$        | splice-block index                |
| $i$                        | $\{1, \dots, n_b\}$                | pseudobulk index within block $b$ |
| $p$                        | $\{1, \dots, P\}$                  | primer-protocol index             |
| $h$                        | $\{1, \dots, H_{bi}\}$             | cell index within pseudobulk $i$  |
| $t$                        | $\{1, \dots, T_g\}$                | isoform index within gene $g(b)$  |
| $s$                        | $\{1, \dots, S_b\}$                | splice-path index                 |
| $\mathbf{c}_{bip}$         | $\mathbb{N}_0^{E_{bp}}$            | pseudobulk EC counts              |
| $A_{bp}$                   | $\mathbb{R}_+^{E_{bp} \times T_g}$ | weighted EC-by-isoform design     |
| $\boldsymbol{\theta}_{bi}$ | $\mathbb{R}_+^{T_g}$               | isoform abundance                 |
| $C_b$                      | $\{0, 1\}^{S_b \times T_g}$        | isoform-to-path map               |
| $Q_b$                      | $\mathbb{R}^{S_b \times d_b}$      | Helmert ILR basis                 |
| $\boldsymbol{\delta}_{bi}$ | $\mathbb{R}^{d_b}$                 | local path perturbation           |
| $\boldsymbol{\psi}_{bi}$   | $\Delta^{S_b - 1}$                 | path composition                  |
| $\mathbf{y}_{bi}$          | $\mathbb{R}^{d_b}$                 | ILR response                      |
| $V_{bi}$                   | $\mathbb{S}_+^{d_b}$               | ILR inferential covariance        |

Here  $\mathcal{B} \in \mathbb{N}$  is the number of blocks,  $\mathbb{N}_0 = \{0, 1, \dots\}$ ,  $\Delta^{S_b-1} = \{\mathbf{v} \in \mathbb{R}_+^{S_b} : \mathbf{1}'\mathbf{v} = 1\}$ , and  $\mathbb{S}_+^{d_b}$  is the set of symmetric positive-semidefinite  $d_b \times d_b$  matrices. Bold lower-case symbols are vectors, upper-case symbols are matrices,  $\mathbf{1}_k \in \mathbb{R}^k$  is a vector of ones,  $\mathbf{0}_k \in \mathbb{R}^k$  is a vector of zeros,  $I_k \in \mathbb{R}^{k \times k}$  is the identity,  $\odot$  is elementwise multiplication, and  $(\cdot)^+$  is the Moore–Penrose pseudoinverse.

## Helmert basis and ILR coordinates

The simplex has only  $d_b = S_b - 1$  free directions because path proportions sum to one. The Helmert matrix  $Q_b$  is a fixed orthonormal basis for the sum-zero subspace:

$$Q_b' Q_b = I_{d_b}, \quad Q_b' \mathbf{1}_{S_b} = \mathbf{0}_{d_b}.$$

Using one-based row  $r \in \{1, \dots, S_b\}$  and column  $j \in \{1, \dots, d_b\}$ , its entries are

$$(Q_b)_{rj} = \begin{cases} 1/\sqrt{j(j+1)}, & r \leq j, \\ -j/\sqrt{j(j+1)}, & r = j+1, \\ 0, & r > j+1. \end{cases}$$

For any strictly positive  $\psi_{bi} \in \Delta^{S_b-1}$ , the ILR coordinate is  $\mathbf{y}_{bi} = Q_b' \log \psi_{bi} \in \mathbb{R}^{d_b}$ , where the logarithm is elementwise. For  $S_b = 2$ ,  $Q_b = (1, -1)'/\sqrt{2}$  and  $y_{bi} = \log(\psi_{bi1}/\psi_{bi2})/\sqrt{2}$ . Thus a two-path ILR test is a scaled log-odds, or differential-PSI, test.

## Statistical Model

### Paired-primer EC likelihood

For block  $b$ , pseudobulk  $i$ , and primer  $p$ , define  $N_{bip} = \mathbf{1}_{E_{bp}}' \mathbf{c}_{bip} \in \mathbb{N}_0$ ,  $\mathbf{q}_{bip} = A_{bp} \boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{E_{bp}}$ , and  $Z_{bip} = \mathbf{1}_{E_{bp}}' \mathbf{q}_{bip} \in \mathbb{R}_+$ . Conditional on the primer-specific UMI total,

$$\mathbf{c}_{bip} \mid N_{bip}, \boldsymbol{\theta}_{bi} \sim \text{Multinomial}(N_{bip}, \mathbf{q}_{bip}/Z_{bip}).$$

The oligo(dT) and random-hexamer protocols use separate  $A_{bp}$  matrices and likelihood terms. Their different isoform exposures therefore model primer-specific positional coverage without forcing a common bias profile.

The genome-wide hierarchical model supplies  $\boldsymbol{\theta}_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$ . Let  $s_b(t) \in \{1, \dots, S_b\}$  map isoform  $t \in \mathcal{I}_b^{\text{sp}}$  to its path. The baseline spliced-isoform mass is the scalar  $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \theta_{bit}^{(0)} \in \mathbb{R}_+$ . For  $t \in \mathcal{I}_b^{\text{sp}}$ , local refinement uses

$$\theta_{bit}(\boldsymbol{\delta}_{bi}) = \omega_{bi}^{\text{sp}} \frac{\theta_{bit}^{(0)} \exp\{(Q_b)_{s_b(t), \cdot} \boldsymbol{\delta}_{bi}\}}{\sum_{u \in \mathcal{I}_b^{\text{sp}}} \theta_{biu}^{(0)} \exp\{(Q_b)_{s_b(u), \cdot} \boldsymbol{\delta}_{bi}\}},$$

where  $(Q_b)_{s, \cdot} \in \mathbb{R}^{1 \times d_b}$ . Nuisance-isoform abundances are unchanged. This preserves total spliced mass and the relative isoform mixture within each path while estimating only  $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$ .

### Path response and inferential covariance

Each column of  $C_b \in \{0, 1\}^{S_b \times T_g}$  corresponding to  $t \in \mathcal{I}_b^{\text{sp}}$  contains one 1 in row  $s_b(t)$ ; nuisance isoform columns are zero. Define

$$\mathbf{a}_{bi} = C_b \boldsymbol{\theta}_{bi} \in \mathbb{R}_+^{S_b}, \quad \boldsymbol{\psi}_{bi} = \frac{\mathbf{a}_{bi}}{\mathbf{1}_{S_b}' \mathbf{a}_{bi}} \in \Delta^{S_b-1}, \quad \mathbf{y}_{bi} = Q_b' \log \boldsymbol{\psi}_{bi} \in \mathbb{R}^{d_b}.$$

The perturbation parameterization implies  $\mathbf{y}_{bi} = \mathbf{y}_{bi}^{(0)} + \boldsymbol{\delta}_{bi}$ , where  $\mathbf{y}_{bi}^{(0)} \in \mathbb{R}^{d_b}$  is computed from the baseline.

Let  $I_{\delta,bi} \in \mathbb{S}_+^{d_b}$  be the expected Fisher information of the summed paired-primer log likelihood with respect to  $\boldsymbol{\delta}_{bi}$ . The conditional covariance is  $V_{bi}^{\text{cond}} = I_{\delta,bi}^{-1} \in \mathbb{S}_+^{d_b}$  when the information is positive definite. It conditions on the fitted total spliced mass, the isoform mixture within each path, and every nuisance-isoform abundance. Its only free directions are the  $d_b$  path perturbations.

### Full-isoform nuisance (profile) covariance

The profile calculation is a covariance sensitivity analysis, not a second estimate of path usage. It retains the same  $\hat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{T_g}$  obtained from the constrained local fit, but asks how uncertain its path ILRs would be if all  $T_g$  isoform log-abundances were locally free. Equivalently, after a smooth reparameterization into path ILRs and  $T_g - d_b$  nuisance coordinates, it approximates the path-ILR block of the inverse Fisher information after profiling the nuisance coordinates. No unconstrained isoform likelihood is optimized, so *profile* below means this local expected-Fisher approximation rather than an exact profile-likelihood fit.

Set  $\boldsymbol{\eta}_{bi} = \log \hat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}^{T_g}$ ,  $D_{bi} = \text{diag}(\hat{\boldsymbol{\theta}}_{bi}) \in \mathbb{R}^{T_g \times T_g}$ ,  $\mathbf{d}_{bp} = A'_{bp} \mathbf{1}_{E_{bp}} \in \mathbb{R}^{T_g}$ , and  $\hat{\mathbf{q}}_{bip} = A_{bp} \hat{\boldsymbol{\theta}}_{bi} \in \mathbb{R}_+^{E_{bp}}$ ,  $\hat{Z}_{bip} = \mathbf{1}_{E_{bp}}' \hat{\mathbf{q}}_{bip} \in \mathbb{R}_+$ , and  $\mathbf{r}_{0,bip} = (\mathbf{d}_{bp} \odot \hat{\boldsymbol{\theta}}_{bi}) / \hat{Z}_{bip} \in \mathbb{R}^{T_g}$ . The primer-specific expected information is

$$I_{\eta,bip} = N_{bip} \left[ \frac{D_{bi} A'_{bp} \text{diag}(\hat{\mathbf{q}}_{bip}^{-1}) A_{bp} D_{bi}}{\hat{Z}_{bip}} - \mathbf{r}_{0,bip} \mathbf{r}_{0,bip}' \right] \in \mathbb{S}_+^{T_g}.$$

The paired-primer information is  $I_{\eta,bi} = \sum_{p=1}^P I_{\eta,bip} \in \mathbb{S}_+^{T_g}$ .

The covariance must be propagated through the isoform-to-path map. Let  $\omega_{bi}^{\text{sp}} = \sum_{t \in \mathcal{I}_b^{\text{sp}}} \hat{\theta}_{bit} \in \mathbb{R}_+$  and  $a_{bis} = \sum_{t: s_b(t)=s} \hat{\theta}_{bit} \in \mathbb{R}_+$ . Define  $R_{bi} = \partial \log \psi_{bi} / \partial \boldsymbol{\eta}'_{bi} \in \mathbb{R}^{S_b \times T_g}$  entrywise by

$$(R_{bi})_{st} = \begin{cases} \mathbb{I}\{s_b(t) = s\} \hat{\theta}_{bit} / a_{bis} - \hat{\theta}_{bit} / \omega_{bi}^{\text{sp}}, & t \in \mathcal{I}_b^{\text{sp}}, \\ 0, & t \notin \mathcal{I}_b^{\text{sp}}. \end{cases}$$

Thus

$$J_{bi} = \frac{\partial \mathbf{y}_{bi}}{\partial \boldsymbol{\eta}'_{bi}} = Q'_b R_{bi} \in \mathbb{R}^{d_b \times T_g}.$$

The multinomial likelihood necessarily has a global-scale null direction, and EC ambiguity can create additional null directions. Write the symmetric eigendecomposition

$$I_{\eta,bi} = U_{bi} \text{diag}(\boldsymbol{\lambda}_{bi}) U_{bi}',$$

where  $U_{bi} \in \mathbb{R}^{T_g \times T_g}$  is orthogonal and  $\boldsymbol{\lambda}_{bi} \in \mathbb{R}^{T_g}$ . An eigenvalue is treated as positive when  $\lambda_{bit} > 10^{-8} \max_u |\lambda_{biu}|$ . Let  $\mathcal{P}_{bi} \subseteq \{1, \dots, T_g\}$  index these positive eigenvalues, let  $\mathcal{N}_{bi}$  index the remainder, and let  $U_{bi,+} \in \mathbb{R}^{T_g \times |\mathcal{P}_{bi}|}$  and  $U_{bi,0} \in \mathbb{R}^{T_g \times |\mathcal{N}_{bi}|}$  contain the corresponding eigenvectors.

The path response is identifiable only if its Jacobian annihilates every information-null direction. The implementation requires

$$\|J_{bi} U_{bi,0}\|_F \leq 10^{-9} \max\{\|J_{bi}\|_F, 1\}.$$

If this fails,  $V_{bi}^{\text{prof}}$  is set to an infinite matrix and the estimate is excluded from profile testing. Otherwise,

$$V_{bi}^{\text{prof}} = J_{bi} U_{bi,+} \text{diag}(\boldsymbol{\lambda}_{bi,+}^{-1}) U_{bi,+}' J_{bi}' = J_{bi} I_{\eta,bi}^+ J_{bi}' \in \mathbb{S}_+^{d_b}.$$

Unlike  $V_{bi}^{\text{cond}}$ , this covariance allows EC-supported changes in within-path isoform mixtures, spliced mass, and nuisance isoforms to absorb information that would otherwise identify a path contrast. It is therefore the more conservative sensitivity calculation when those nuisance directions are weakly identified.

### Cell-resolved likelihood

For cell  $h$  in pseudobulk  $i$ , let  $\mathbf{c}_{bihp} \in \mathbb{N}_0^{E_{bp}}$ ,  $N_{bihp} = \mathbf{1}'_{E_{bp}} \mathbf{c}_{bihp} \in \mathbb{N}_0$ , and  $\boldsymbol{\theta}_{bih}^{(0)} \in \mathbb{R}_+^{T_g}$  be its EC counts, total, and baseline. Every cell shares the same pseudobulk perturbation  $\boldsymbol{\delta}_{bi} \in \mathbb{R}^{d_b}$ , but its isoform vector  $\boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi}) \in \mathbb{R}_+^{T_g}$  is normalized using its own baseline. The objective is the scalar

$$\ell_{bi}(\boldsymbol{\delta}_{bi}) = \sum_{h=1}^{H_{bi}} \sum_{p=1}^P \log \text{Multinomial} \left( \mathbf{c}_{bihp}; N_{bihp}, \frac{A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})}{\mathbf{1}'_{E_{bp}} A_{bp} \boldsymbol{\theta}_{bih}(\boldsymbol{\delta}_{bi})} \right).$$

Thus cells contribute assignment information but do not become biological replicates. Let  $U_{bih} \in \mathbb{R}_+$  be the cell's total UMI count and  $g_{bih} \in [0, 1]$  its decoded proportion for the block's gene. The scalar weight is  $w_{bih} = U_{bih} g_{bih}$ , and the reported pseudobulk composition is

$$\boldsymbol{\psi}_{bi} = \frac{\sum_{h=1}^{H_{bi}} w_{bih} \boldsymbol{\psi}_{bih}}{\sum_{h=1}^{H_{bi}} w_{bih}} \in \Delta^{S_b-1}.$$

Cell- and primer-specific Fisher information matrices in  $\mathbb{S}_+^{d_b}$  are summed, inverted, and propagated through this weighted average to obtain  $V_{bi} \in \mathbb{S}_+^{d_b}$ .

### Effective counts and compositional regression

Treating all gene UMIs as known path assignments is overconfident when ECs are ambiguous. For each fitted  $\boldsymbol{\psi}_{bi}$ , define the unit-multinomial ILR covariance

$$G_{bi} = Q_b' \text{diag}(\boldsymbol{\psi}_{bi}^{-1}) Q_b \in \mathbb{S}_+^{d_b}.$$

The scalar effective count matches  $G_{bi}/n$  to the EC-derived  $V_{bi}^{\text{cond}}$  in Frobenius inner product  $\langle U, W \rangle_F = \text{tr}(U'W)$ , for  $U, W \in \mathbb{R}^{d_b \times d_b}$ :

$$n_{bi}^{\text{eff}} = \min \left\{ N_{bi}^{\text{gene}}, \frac{\langle G_{bi}, G_{bi} \rangle_F}{\langle G_{bi}, V_{bi}^{\text{cond}} \rangle_F} \right\} \in \mathbb{R}_+,$$

where  $N_{bi}^{\text{gene}} \in \mathbb{N}_0$  is the gene UMI count. Fractional path counts are  $\mathbf{x}_{bi} = n_{bi}^{\text{eff}} \boldsymbol{\psi}_{bi} \in \mathbb{R}_+^{S_b}$ , with  $\mathbf{1}'_{S_b} \mathbf{x}_{bi} = n_{bi}^{\text{eff}}$ .

For one block, stack  $n = n_b \in \mathbb{N}$  observations in a design  $X \in \mathbb{R}^{n \times k}$ , where  $k \in \mathbb{N}$ , and coefficient matrix  $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$ . The mean composition  $\boldsymbol{\mu}_{bi} \in \Delta^{S_b-1}$  is defined by reference-path logits

$$\log \frac{\mu_{bis}}{\mu_{biS_b}} = (X \mathbf{B}_b)_{is}, \quad s = 1, \dots, d_b.$$

The generalized Dirichlet-multinomial likelihood, extended to fractional counts through gamma functions, is

$$\mathbf{x}_{bi} \mid n_{bi}^{\text{eff}} \sim \text{DirichletMultinomial} \left( n_{bi}^{\text{eff}}, \kappa_b \boldsymbol{\mu}_{bi} \right),$$

where  $\kappa_b \in \mathbb{R}_+ \cup \{\infty\}$  is a block-level concentration. Larger  $\kappa_b$  means less overdispersion; the endpoint  $\kappa_b = \infty$  is exactly multinomial.

For nested designs  $X_0 \in \mathbb{R}^{n \times k_0}$  and  $X_1 \in \mathbb{R}^{n \times k_1}$ , with  $k_1 \geq k_0$  and the first  $k_0, k_1 \in \mathbb{N}$ , and the first  $k_0$  columns of  $X_1$  equal to  $X_0$ , let  $\widehat{\ell}_0, \widehat{\ell}_1 \in \mathbb{R}$  be maximized scalar log likelihoods. The statistic and nominal degrees of freedom are

$$\Lambda_b = 2(\widehat{\ell}_1 - \widehat{\ell}_0) \in \mathbb{R}_+, \quad \nu_b = (k_1 - k_0)d_b \in \mathbb{N}_0.$$

The fixed-concentration analysis estimates  $\widehat{\kappa}_{0b} \in \mathbb{R}_+ \cup \{\infty\}$  under  $X_0$  and uses it under  $X_1$ . The free-concentration sensitivity estimates  $\widehat{\kappa}_{1b} \in \mathbb{R}_+ \cup \{\infty\}$  under  $X_1$ . In the latter case both models contain one concentration nuisance parameter, so  $\nu_b$  is unchanged. Both finite and exact multinomial endpoints are evaluated.

## Gaussian mouse-replicate regression

For a condition test within one block and cell type, stack  $\widehat{\mathbf{y}}_{bi} \in \mathbb{R}^{d_b}$  over  $n \in \mathbb{N}$  mice. Let  $X_b \in \mathbb{R}^{n \times k}$ ,  $k \in \mathbb{N}$ , contain an intercept and condition indicators,  $\mathbf{B}_b^{\text{GLS}} \in \mathbb{R}^{k \times d_b}$  be regression coefficients, and  $\epsilon_{bi} \in \mathbb{R}^{d_b}$  be residuals:

$$\widehat{\mathbf{y}}_{bi} = (X_{b,i} \mathbf{B}_b^{\text{GLS}})' + \epsilon_{bi}, \quad \text{Var}(\epsilon_{bi}) = V_{bi} + \tau_b^2 I_{d_b} \in \mathbb{S}_+^{d_b}.$$

Here  $X_{b,i} \in \mathbb{R}^{1 \times k}$ , and  $\tau_b^2 \in \mathbb{R}_+$  is residual between-mouse variance estimated by REML under the null. If  $r_b \in \mathbb{N}$  non-reference condition columns are tested, their coefficients form  $\beta_b^{\text{test}} \in \mathbb{R}^{r_b d_b}$  with covariance  $\Sigma_b^{\text{test}} \in \mathbb{S}_+^{r_b d_b}$ . The Wald statistic is

$$W_b = (\beta_b^{\text{test}})' (\Sigma_b^{\text{test}})^{-1} \beta_b^{\text{test}} \in \mathbb{R}_+, \quad \nu_b^{\text{GLS}} = r_b d_b.$$

## Paired cell-type test and permutations

For a cell-type test, let  $M_b, L_b \in \mathbb{N}$  be the numbers of represented biological mice and cell types, respectively. The null design  $X_{0b} \in \{0, 1\}^{n_b \times M_b}$  contains an intercept-coded set of mouse fixed effects. The alternative  $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$  appends treatment-coded cell-type indicators. Consequently  $k_0 = M_b$ ,  $k_1 = M_b + L_b - 1$ , and  $\nu_b = (L_b - 1)d_b$ . The tested coefficient block is  $\mathbf{B}_b^{\text{cell}} \in \mathbb{R}^{(L_b - 1) \times d_b}$ , and

$$H_{0b} : \mathbf{B}_b^{\text{cell}} = 0 \quad \text{versus} \quad H_{1b} : \mathbf{B}_b^{\text{cell}} \neq 0.$$

Mouse effects absorb condition and every shift shared by cell types from the same mouse.

For  $R = 500 \in \mathbb{N}$  permutations and  $r \in \{1, \dots, R\}$ , cell-type labels are shuffled only among rows from the same mouse, producing  $X_{1b}^{(r)} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$  and statistic  $\Lambda_b^{(r)} \in \mathbb{R}_+$ . Let  $\varepsilon_b = 10^{-6} \max(1, |\Lambda_b|) \in \mathbb{R}_+$ . The finite-permutation p-value is

$$p_b = \frac{1 + \sum_{r=1}^R \mathbb{1}\{\Lambda_b^{(r)} \geq \Lambda_b - \varepsilon_b\}}{R + 1} \in (0, 1], \quad R = 500.$$

The null coefficients and  $\widehat{\kappa}_{0b}$  are reused, but each permuted alternative is optimized. The tolerance counts numerically equal likelihood optima as ties. Benjamini–Hochberg correction is applied across canonical partitions with at least 30 pseudobulk rows.

## Joint condition tests

ACAT combines  $J \in \mathbb{N}$  component p-values  $p_j \in (0, 1)$ ,  $j = 1, \dots, J$ , into the scalar

$$p_{\text{ACAT}} = \frac{1}{2} - \frac{1}{\pi} \arctan \left[ \frac{1}{J} \sum_{j=1}^J \tan\{\pi(1/2 - p_j)\} \right] \in (0, 1).$$

One level combines cell-type-specific p-values for a path partition; a second combines distinct partitions within a gene.

The shared-effect Gaussian model for block  $b$  uses  $M \in \mathbb{N}$  mice indexed by  $m \in \{1, \dots, M\}$ , and  $L_m \in \mathbb{N}$  cell types for mouse  $m$ , indexed by  $c \in \{1, \dots, L_m\}$ . Its response is  $\mathbf{y}_{bmc} \in \mathbb{R}^{d_b}$ , design row  $X_{bmc} \in \mathbb{R}^{1 \times k}$ , where  $k \in \mathbb{N}$ , coefficient matrix  $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$ , mouse random effect  $\mathbf{u}_{bm} \in \mathbb{R}^{d_b}$ , and residual  $\mathbf{e}_{bmc} \in \mathbb{R}^{d_b}$ :

$$\mathbf{y}_{bmc} = (X_{bmc}\mathbf{B}_b)' + \mathbf{u}_{bm} + \mathbf{e}_{bmc}.$$

The random terms satisfy  $E(\mathbf{u}_{bm}) = E(\mathbf{e}_{bmc}) = \mathbf{0}_{d_b}$ ,  $\text{Var}(\mathbf{u}_{bm}) = \tau_{\text{mouse}}^2 I_{d_b}$ , and  $\text{Var}(\mathbf{e}_{bmc}) = V_{bmc} + \tau_{\text{resid}}^2 I_{d_b}$ . They are independent across mice, and residuals are independent across cell types conditional on  $\mathbf{u}_{bm}$ . Hence, for  $c \neq c'$ ,

$$\text{Cov}(\mathbf{y}_{bmc}, \mathbf{y}_{bmc'}) = \tau_{\text{mouse}}^2 I_{d_b}, \quad \text{Var}(\mathbf{y}_{bmc}) = V_{bmc} + (\tau_{\text{mouse}}^2 + \tau_{\text{resid}}^2) I_{d_b},$$

where  $V_{bmc} \in \mathbb{S}_+^{d_b}$ , and  $\tau_{\text{mouse}}^2, \tau_{\text{resid}}^2 \in \mathbb{R}_+$  are shared-mouse and residual variance components estimated by two-component REML under the no-condition model. Condition labels are permuted once per mouse and applied to all its cell types.

## Multinomial-logit GLMM

The formal generalized linear mixed model uses the same shared-condition design as the joint Gaussian model but retains the multinomial response. Let  $\tilde{\mathbf{x}}_{bmc} \in \mathbb{N}_0^{S_b}$  be integer path counts with  $\tilde{n}_{bmc} = \mathbf{1}_{S_b}' \tilde{\mathbf{x}}_{bmc} \in \mathbb{N}$ . They are obtained from the fractional effective counts  $\mathbf{x}_{bmc}$  by rounding  $n_{bmc}^{\text{eff}}$  to the nearest positive integer and distributing the remaining molecules to paths by largest fractional remainder. This makes the conditional response a proper multinomial while changing each effective total by at most one.

For design row  $X_{bmc} \in \mathbb{R}^{1 \times k}$ , fixed coefficients  $\mathbf{B}_b \in \mathbb{R}^{k \times d_b}$ , and mouse random intercept  $\mathbf{u}_{bm} \in \mathbb{R}^{d_b}$ , define

$$\boldsymbol{\xi}_{bmc} = X_{bmc}\mathbf{B}_b + \mathbf{u}_{bm}' \in \mathbb{R}^{1 \times d_b}, \quad \boldsymbol{\mu}_{bmc} = \text{softmax}\{(\boldsymbol{\xi}_{bmc}, 0)\} \in \Delta^{S_b-1}.$$

The final path is the reference category. The model is

$$\tilde{\mathbf{x}}_{bmc} \mid \mathbf{u}_{bm} \sim \text{Multinomial}(\tilde{n}_{bmc}, \boldsymbol{\mu}_{bmc}), \quad \mathbf{u}_{bm} \sim N(\mathbf{0}_{d_b}, \sigma_b^2 I_{d_b}),$$

where  $\sigma_b^2 \in \mathbb{R}_+$  is re-estimated under both null and alternative. An isotropic random-effect covariance is used because some blocks have few mice; an unstructured  $d_b \times d_b$  covariance would introduce  $d_b(d_b + 1)/2$  variance parameters per block.

Let  $\ell_{bm}(\mathbf{u}; \mathbf{B}_b, \sigma_b) \in \mathbb{R}$  be mouse  $m$ 's conditional multinomial log likelihood plus its Gaussian log density, and let

$$\hat{\mathbf{u}}_{bm} = \arg \max_{\mathbf{u} \in \mathbb{R}^{d_b}} \ell_{bm}(\mathbf{u}; \mathbf{B}_b, \sigma_b), \quad H_{bm} = - \frac{\partial^2 \ell_{bm}}{\partial \mathbf{u} \partial \mathbf{u}'} \Big|_{\hat{\mathbf{u}}_{bm}} \in \mathbb{S}_+^{d_b}.$$

The Laplace log marginal likelihood, up to constants common to nested models, is

$$\ell_b^{\text{Lap}}(\mathbf{B}_b, \sigma_b) = \sum_{m=1}^M \left[ \ell_{bm}(\hat{\mathbf{u}}_{bm}; \mathbf{B}_b, \sigma_b) + \frac{d_b}{2} \log(2\pi) - \frac{1}{2} \log |H_{bm}| \right] \in \mathbb{R}.$$

The implementation differentiates this approximation through 30 damped Newton updates for every  $\hat{\mathbf{u}}_{bm}$ . The null contains cell-type intercepts; the alternative appends  $r_b$  condition columns. The likelihood-ratio statistic has nominal degrees of freedom  $r_b d_b$ .

## Liang–Zeger multinomial GEE

GEE specifies only the marginal mean and first two moments. It can therefore use the fractional effective counts directly. Define  $\mathbf{p}_{bmc} = \mathbf{x}_{bmc}/n_{bmc}^{\text{eff}} \in \Delta^{S_b-1}$ , and let  $\mathbf{p}_{bmc,-} \in \mathbb{R}^{d_b}$  omit the reference path. Its marginal mean uses the same reference-logit regression,

$$\boldsymbol{\mu}_{bmc} = \text{softmax}\{(X_{bmc}\mathbf{B}_b, 0)\}, \quad G_{bmc} = \text{diag}(\boldsymbol{\mu}_{bmc,-}) - \boldsymbol{\mu}_{bmc,-}\boldsymbol{\mu}_{bmc,-}' \in \mathbb{S}_+^{d_b}.$$

The working covariance of one proportion vector is  $A_{bmc} = G_{bmc}/n_{bmc}^{\text{eff}} \in \mathbb{S}_+^{d_b}$ . Let  $L_{bmc} \in \mathbb{R}^{d_b \times d_b}$  be its symmetric square root, and let

$$D_{bmc} = \frac{\partial \boldsymbol{\mu}_{bmc,-}}{\partial \text{vec}(\mathbf{B}_b)'} = [X_{bmc,1}G_{bmc} \ \cdots \ X_{bmc,k}G_{bmc}] \in \mathbb{R}^{d_b \times kd_b}.$$

For mouse  $m$ , stack its  $q_m \in \mathbb{N}$  observed cell types into  $\mathbf{p}_{bm,-} \in \mathbb{R}^{q_m d_b}$ , mean  $\boldsymbol{\mu}_{bm,-} \in \mathbb{R}^{q_m d_b}$ , and derivative  $D_{bm} \in \mathbb{R}^{q_m d_b \times kd_b}$ . With  $L_{bm} = \text{blockdiag}_c(L_{bmc})$  and exchangeable  $R_m(\alpha_b) \in \mathbb{R}^{q_m \times q_m}$ , the Liang–Zeger working covariance is

$$W_{bm} = \phi_b L_{bm} \{R_m(\alpha_b) \otimes I_{d_b}\} L_{bm}' \in \mathbb{S}_+^{q_m d_b}.$$

The scale  $\phi_b \in \mathbb{R}_+$  and working correlation  $\alpha_b \in (-1/(q_{\max} - 1), 1)$  are updated from whitened Pearson residuals. Coefficients solve

$$\sum_{m=1}^M D_{bm}' W_{bm}^{-1} (\mathbf{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) = \mathbf{0}_{kd_b}.$$

If  $\mathbf{U}_{bm} = D_{bm}' W_{bm}^{-1} (\mathbf{p}_{bm,-} - \boldsymbol{\mu}_{bm,-}) \in \mathbb{R}^{kd_b}$  and  $\mathcal{H}_b = \sum_m D_{bm}' W_{bm}^{-1} D_{bm} \in \mathbb{S}_+^{kd_b}$ , the canonical cluster-robust covariance is

$$\widehat{\text{Var}}\{\text{vec}(\hat{\mathbf{B}}_b)\} = \mathcal{H}_b^+ \left( \sum_{m=1}^M \mathbf{U}_{bm} \mathbf{U}_{bm}' \right) \mathcal{H}_b^+ \in \mathbb{S}_+^{kd_b}.$$

The condition coefficients are tested by a robust Wald statistic. No small-sample sandwich correction is imposed; synchronized mouse-level permutations assess whether the canonical large- $M$  reference is adequate.

## Matched power benchmark

Power is evaluated only for partitions represented by exactly two conditions. For replicate  $r$ , condition labels are first permuted among mice. Let  $z_{bmc}^{(r)} \in \{0, 1\}$  be the resulting non-reference condition indicator and let  $\mathbf{v}_b \in \mathbb{R}^{d_b}$  be a reproducible random unit vector. For ILR effect  $\gamma \in \{0.25, 0.5\}$ , define

$$\mathbf{y}_{bmc}^{(r,\gamma)} = \mathbf{y}_{bmc} + \gamma z_{bmc}^{(r)} \mathbf{v}_b, \quad \boldsymbol{\psi}_{bmc}^{(r,\gamma)} = \text{softmax}\{Q_b \mathbf{y}_{bmc}^{(r,\gamma)}\}.$$

The shifted effective counts are  $n_{bmc}^{\text{eff}} \boldsymbol{\psi}_{bmc}^{(r,\gamma)}$ . Thus every method sees the same imposed ILR effect, observed effective depth, cell-type pattern, and mouse clustering. For each simulated alternative, null-calibrated power uses the finite-permutation rank of its statistic among synchronized null statistics for the same block and method. This event-matched calibration preserves dimension, depth, and cluster structure.



## Algorithms

All algorithms operate independently over canonical blocks unless a joint gene-level combination is stated. “Fit” below means bounded L-BFGS with analytic gradients and a fourfold iteration-limit retry after a warm start from the first endpoint.

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### Algorithm 1 Construct splice blocks and path maps

---

**Require:**  $K_g$  annotated spliced isoforms with exon chains; retained set  $\mathcal{I}_g$ ,  $|\mathcal{I}_g| = T_g$

**Ensure:** Blocks  $b$ , maps  $C_b \in \{0, 1\}^{S_b \times T_g}$ , and bases  $Q_b \in \mathbb{R}^{S_b \times d_b}$

- 1: Split the gene at every annotated exon boundary into atomic intervals.
  - 2: Merge adjacent intervals exonic in every annotated isoform into constitutive anchors.
  - 3: Form internal blocks between consecutive anchors and terminal blocks using pseudo-anchors.
  - 4: **for all** candidate blocks  $b$  **do**
  - 5:     Record each annotated isoform’s ordered intervening exon chain.
  - 6:     Assign identical chains to one path and restrict to  $\mathcal{I}_b^{\text{sp}} \subseteq \mathcal{I}_g$ .
  - 7:     Remap represented path labels to  $\{1, \dots, S_b\}$ .
  - 8:     **if**  $S_b < 2$  **then**
  - 9:         discard  $b$
  - 10:    **else**
  - 11:       Set  $(C_b)_{s_b(t), t} = 1$  for  $t \in \mathcal{I}_b^{\text{sp}}$ ; leave nuisance columns zero.
  - 12:       Construct the Helmert basis  $Q_b$ .
  - 13:    **end if**
  - 14: **end for**
  - 15: Canonicalize the unordered isoform partition and retain one representative of equivalent blocks.
- 

---

### Algorithm 2 Fit one pseudobulk’s local path composition

---

**Require:** Counts  $c_{bip} \in \mathbb{N}_0^{E_{bp}}$ , designs  $A_{bp} \in \mathbb{R}_+^{E_{bp} \times T_g}$ , baseline  $\theta_{bi}^{(0)} \in \mathbb{R}_+^{T_g}$ ,  $C_b$ , and  $Q_b$

**Ensure:**  $\hat{\theta}_{bi}$ ,  $\hat{\psi}_{bi}$ ,  $\hat{y}_{bi}$ , and  $V_{bi}^{\text{cond}}$

- 1: Initialize  $\delta_{bi} \leftarrow \mathbf{0}_{d_b}$ .
  - 2: **if**  $S_b = 2$  **then**
  - 3:     Evaluate the fixed log-ratio grid and retain its best point.
  - 4: **end if**
  - 5: **function** OBJECTIVE( $\delta$ )
  - 6:     Form  $\theta_{bi}(\delta)$ , preserving spliced mass and within-path isoform mixtures.
  - 7:     **return** the negative sum of the  $P = 2$  multinomial log likelihoods and its analytic gradient.
  - 8: **end function**
  - 9: Minimize OBJECTIVE( $\delta$ ) by bounded L-BFGS on  $[-20, 20]^{d_b}$ .
  - 10: Set  $\hat{\theta}_{bi} \leftarrow \theta_{bi}(\hat{\delta}_{bi})$ .
  - 11: Set  $\hat{\psi}_{bi} \leftarrow C_b \hat{\theta}_{bi} / (\mathbf{1}_{S_b}' C_b \hat{\theta}_{bi})$ .
  - 12: Set  $\hat{y}_{bi} \leftarrow Q_b' \log \hat{\psi}_{bi}$ .
  - 13: Compute  $I_{\delta, bi}$  by summing primer expected information.
  - 14: Apply the eigenvalue/null-space check to  $(I_{\delta, bi}, I_{d_b})$ ; if identifiable, set  $V_{bi}^{\text{cond}} = I_{\delta, bi}^{-1}$ , otherwise set it to infinity.
-

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**Algorithm 3** Compute full-isoform nuisance (profile) covariance

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**Require:**  $\hat{\theta}_{bi} \in \mathbb{R}_+^{T_g}$ ,  $A_{bp}$ ,  $N_{bip}$ ,  $C_b$ ,  $Q_b$ , relative threshold  $r_{\text{eig}} = 10^{-8}$ , and null tolerance  $a_{\text{null}} = 10^{-9}$

**Ensure:**  $V_{bi}^{\text{prof}} \in \mathbb{S}_+^{d_b}$  and an identifiability flag

- 1: **for**  $p = 1, \dots, P$  **do**
  - 2:   Compute  $\hat{q}_{bip}$ ,  $\hat{Z}_{bip}$ , and  $I_{\eta,bip} \in \mathbb{S}_+^{T_g}$ .
  - 3: **end for**
  - 4: Set  $I_{\eta,bi} \leftarrow \sum_p I_{\eta,bip}$ .
  - 5: Compute path masses  $a_{bis}$ , spliced mass  $\omega_{bi}^{\text{sp}}$ , and  $R_{bi} = \partial \log \psi_{bi} / \partial \eta'_{bi}$ .
  - 6: Set  $J_{bi} \leftarrow Q'_b R_{bi}$ .
  - 7: Eigendecompose  $I_{\eta,bi} = U_{bi} \text{diag}(\lambda_{bi}) U'_{bi}$ .
  - 8: Put  $t$  in  $\mathcal{P}_{bi}$  when  $\lambda_{bit} > r_{\text{eig}} \max_u |\lambda_{biu}|$ , and put all other indices in  $\mathcal{N}_{bi}$ .
  - 9: **if**  $\|J_{bi} U_{bi,0}\|_F > a_{\text{null}} \max\{\|J_{bi}\|_F, 1\}$  **then**
  - 10:   Set  $V_{bi}^{\text{prof}} \leftarrow \infty_{d_b \times d_b}$  and mark the response unidentified.
  - 11: **else**
  - 12:   Set  $V_{bi}^{\text{prof}} \leftarrow J_{bi} U_{bi,+} \text{diag}(\lambda_{bi,+}^{-1}) U'_{bi,+} J'_{bi}$ .
  - 13:   Mark the response identifiable.
  - 14: **end if**
- 

---

**Algorithm 4** Fit a cell-resolved pseudobulk path composition

---

**Require:** Cell EC vectors  $c_{bihp}$ , baselines  $\theta_{bih}^{(0)}$ , weights  $w_{bih}$ , and  $A_{bp}$ ,  $C_b$ ,  $Q_b$

**Ensure:** One pseudobulk response  $\hat{y}_{bi}$  and covariance  $V_{bi}$

- 1: Allocate one shared  $\delta_{bi} \in \mathbb{R}^{d_b}$ .
  - 2: **for all** cells  $h$  and primers  $p$  **do**
  - 3:   Perturb the cell's baseline, normalize its EC probabilities, and add its multinomial log likelihood and gradient.
  - 4: **end for**
  - 5: Optimize the summed objective over  $\delta_{bi}$ .
  - 6: Compute each fitted  $\hat{\psi}_{bih}$  and their normalized  $w_{bih}$ -weighted mean  $\hat{\psi}_{bi}$ .
  - 7: Sum cell-by-primer Fisher matrices and propagate their inverse through the weighted composition response to obtain  $V_{bi}$ .
  - 8: Set  $\hat{y}_{bi} = Q'_b \log \hat{\psi}_{bi}$ .
-

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**Algorithm 5** Gaussian condition or paired cell-type test

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**Require:** Responses  $\hat{\mathbf{y}}_{bi}$ , covariances  $V_{bi}$ , mouse, condition, and cell-type labels; test type

**Ensure:** One Wald p-value for block  $b$

- 1: **if** testing condition within a cell type **then**
  - 2:     Retain conditions represented by at least three mice.
  - 3:     Build full design  $X_b \in \mathbb{R}^{n \times k}$  and reduced design  $X_{0b} \in \mathbb{R}^{n \times (k-r_b)}$ .
  - 4: **else**
  - 5:     Build mouse-effect design  $X_{0b} \in \{0, 1\}^{n_b \times M_b}$ .
  - 6:     Append  $L_b - 1$  cell-type columns to obtain  $X_{1b} \in \mathbb{R}^{n_b \times (M_b + L_b - 1)}$ .
  - 7: **end if**
  - 8:     Require full column rank in both designs.
  - 9:     Estimate  $\tau_b^2$  by REML under the reduced design using  $V_{bi} + \tau_b^2 I_{d_b}$ .
  - 10:     Hold  $\hat{\tau}_b^2$  fixed and fit the full design by GLS.
  - 11:     Stack the tested coefficients and compute their Wald statistic  $W_b$ .
  - 12:     Compare  $W_b$  with the applicable chi-squared reference.
  - 13:     Calibrate by mouse-level condition permutations or within-mouse cell-type permutations, as appropriate.
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**Algorithm 6** Paired compositional cell-type test

---

**Require:**  $\hat{\psi}_{bi}$ ,  $V_{bi}^{\text{cond}}$ , mouse and cell-type labels, and  $R = 500$

**Ensure:** One exact permutation p-value  $p_b$  per canonical block

- 1:     Keep converged, identifiable estimates satisfying the path floor and the three-mouse-per-cell-type requirement.
  - 2:     Convert each  $(\hat{\psi}_{bi}, V_{bi}^{\text{cond}})$  to  $n_{bi}^{\text{eff}}$  and  $\mathbf{x}_{bi}$ .
  - 3:     Build  $X_{0b}$  from mouse effects and  $X_{1b}$  by appending cell-type effects.
  - 4:     Fit the null coefficients and  $\hat{\kappa}_{0b}$ .
  - 5:     Fit the observed alternative and compute  $\Lambda_b$ .
  - 6:     **if** using fixed concentration **then**
  - 7:         Set  $\kappa_{1b} \leftarrow \hat{\kappa}_{0b}$ .
  - 8:     **else**
  - 9:         Optimize  $\hat{\kappa}_{1b}$  and compare the finite optimum with the exact multinomial endpoint.
  - 10:     **end if**
  - 11:     **for**  $r = 1, \dots, R$  **do**
  - 12:         Shuffle cell-type labels within each mouse to form  $X_{1b}^{(r)}$ .
  - 13:         Reuse the null fit, refit the alternative, and store  $\Lambda_b^{(r)}$ .
  - 14:     **end for**
  - 15:     Compute the tolerance-aware finite-permutation rank  $p_b$ .
  - 16:     Require  $n_b \geq 30$ , then apply BH across canonical blocks.
-

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**Algorithm 7** Liang–Zeger multinomial GEE condition test

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**Require:** Fractional counts  $\mathbf{x}_{bmc}$ , design  $X_b$ , mouse labels, and tested condition columns

**Ensure:** Robust Wald statistic and p-value

- 1: Initialize  $\mathbf{B}_b$  from independent multinomial regression and set  $\alpha_b \leftarrow 0$ ,  $\phi_b \leftarrow 1$ .
  - 2: **repeat**
  - 3:   Compute  $\boldsymbol{\mu}_{bmc}$ ,  $G_{bmc}$ ,  $A_{bmc}$ , and  $D_{bmc}$  for every observation.
  - 4:   Whiten residuals by  $A_{bmc}^{-1/2}$ .
  - 5:   Update  $\phi_b$  from squared Pearson residuals and  $\alpha_b$  from within-mouse Pearson cross-products.
  - 6:   Construct every  $W_{bm}$ , quasi-score  $\mathbf{U}_{bm}$ , and sensitivity matrix  $\mathcal{H}_b$ .
  - 7:   Update  $\text{vec}(\mathbf{B}_b) \leftarrow \text{vec}(\mathbf{B}_b) + \mathcal{H}_b^+ \sum_m \mathbf{U}_{bm}$ .
  - 8:   **until** the maximum coefficient update is at most  $10^{-8}$
  - 9:   Form the cluster sandwich covariance  $\mathcal{H}_b^+ (\sum_m \mathbf{U}_{bm} \mathbf{U}_{bm}') \mathcal{H}_b^+$ .
  - 10:   Compute the robust Wald test for all condition coefficients.
  - 11:   Permute condition once per mouse and repeat the complete GEE fit.
- 

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**Algorithm 8** Laplace multinomial GLMM condition test

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**Require:** Effective counts, nested designs, mouse labels, and random-effect dimension  $d_b$

**Ensure:** Likelihood-ratio statistic and p-value

- 1: Integerize each effective count vector by largest remainder.
  - 2: **for all** null and alternative models **do**
  - 3:   Initialize fixed coefficients from multinomial regression and initialize  $\sigma_b$ .
  - 4:   **repeat**
  - 5:     **for all** mice  $m$  **do**
  - 6:       Find  $\hat{\mathbf{u}}_{bm}$  with 30 damped Newton updates.
  - 7:       Compute random-effect Hessian  $H_{bm}$  and its log determinant.
  - 8:     **end for**
  - 9:     Sum the Laplace log marginal likelihood and differentiate through the Newton updates.
  - 10:    Update  $\mathbf{B}_b$  and  $\log \sigma_b$  by bounded L-BFGS.
  - 11:   **until** the outer gradient tolerance or iteration limit is reached
  - 12: **end for**
  - 13: Compute twice the null-to-alternative Laplace log-likelihood gain.
  - 14: Compare with  $\chi^2_{r_b d_b}$ .
  - 15: For permutations, reuse the fitted null and refit each permuted alternative.
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**Algorithm 9** Matched semi-synthetic power benchmark

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**Require:** Two-condition partitions, effect sizes  $\gamma$ , and replicate count  $R_{\text{power}}$

**Ensure:** Asymptotic and null-calibrated power for each method

- 1: **for all** eligible blocks  $b$  **do**
  - 2:     Draw and fix a unit ILR direction  $\mathbf{v}_b$ .
  - 3:     **for**  $r = 1, \dots, R_{\text{power}}$  **do**
  - 4:         Permute condition labels among mice.
  - 5:         **for all** nonzero effect sizes  $\gamma$  **do**
  - 6:             Set  $\mathbf{y}_{bmc}^{(r,\gamma)} = \mathbf{y}_{bmc} + \gamma z_{bmc}^{(r)} \mathbf{v}_b$ .
  - 7:             Map shifted ILRs to compositions and effective path counts.
  - 8:             Fit GEE, GLMM, clustered GLS, and Dirichlet–multinomial tests.
  - 9:         **end for**
  - 10:     **end for**
  - 11: **end for**
  - 12: Rank each alternative statistic against null statistics for its block and method, using the finite-permutation correction.
  - 13: Report asymptotic and event-matched permutation rejection at 0.05.
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---

**Algorithm 10** Pairwise and dominant-path sensitivity tests

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**Require:** Path estimates and conditional covariances from Algorithm 2

**Ensure:** Calibrated sensitivity tables

- 1: **for all** cell-type pairs **do**
  - 2:     Retain  $M$  mice containing both types.
  - 3:     Use  $X_0 \in \mathbb{R}^{2M \times M}$  for mouse effects and  $X_1 \in \mathbb{R}^{2M \times (M+1)}$  for mouse plus cell type.
  - 4:     Permute the two labels within mouse and correct across all block-by-pair hypotheses.
  - 5: **end for**
  - 6: **for all** blocks  $b$  **do**
  - 7:     Rank paths by  $\sum_i N_{bi}^{\text{gene}} \psi_{bis}$  and select  $\mathcal{S}_b$ , the two largest, without labels.
  - 8:     Construct selector  $E_b$ , two-path basis  $Q_b^{(2)}$ , and  $\mathcal{R}_b = (Q_b^{(2)})' E_b' Q_b$ .
  - 9:     Set  $V_{bi}^{(2)} = \mathcal{R}_b V_{bi} \mathcal{R}_b'$  and renormalize the two retained proportions.
  - 10:     Run Algorithm 6 and record path-floor sensitivity sweeps.
  - 11: **end for**
-

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**Algorithm 11** Joint condition tests

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**Require:** Calibrated block-by-cell-type p-values or ILR responses

**Ensure:** Partition- or gene-level joint p-values

- 1: For ACAT, combine valid cell-type p-values within a canonical partition; optionally combine partition p-values within a gene.
  - 2: As a negative control, fit one shared condition effect while treating cell-type rows as independent.
  - 3: For clustered GLS, construct diagonal covariance blocks  $V_{bmc} + (\tau_{\text{mouse}}^2 + \tau_{\text{resid}}^2)I_{d_b}$  and same-mouse off-diagonal blocks  $\tau_{\text{mouse}}^2 I_{d_b}$ .
  - 4: Estimate  $\tau_{\text{mouse}}^2$  and  $\tau_{\text{resid}}^2$  by REML under the null.
  - 5: Fit the shared condition coefficient by GLS.
  - 6: **for all** mouse-level condition permutations **do**
  - 7:     Apply one permuted condition to all cell types from that mouse and refit the model.
  - 8: **end for**
  - 9: Apply BH to the canonical-partition family; keep negative-control and calibrated results separate.
- 

## Implementation Details

### Splice-block construction

GTF exons are represented by zero-based, half-open intervals. Exon boundaries define atomic intervals; intervals exonic in every annotated spliced isoform are merged into constitutive anchors. Consecutive anchors define a block, and an isoform path is its ordered exon-interval chain between them. Terminal pseudo-anchors include alternative first and last regions. A gene without a shared constitutive region receives one whole-gene block.

Blocks with one path are discarded. Simple two-path blocks include skipped exons, mutually exclusive exons, alternative donors and acceptors, and retained introns. The implementation also supports nested and multi-path blocks. Precursor isoforms remain nuisance isoforms in the EC model but are not assigned to spliced paths.

Different genomic blocks can induce exactly the same partition of annotated isoforms, for example when several exons always occur together. Such blocks have identical statistical responses and must not count as repeated evidence. A canonical partition is the unordered collection of isoform sets assigned to paths. Equivalent blocks are collapsed before FDR or joint testing; all member interval identifiers remain in the output.

### Pseudobulks and local fitting

Cell isoform proportions from the hierarchical decoder are weighted by cell UMI totals and aggregated by mouse, condition, and cell type. Pseudobulks must contain at least 20 cells and  $10^5$  total UMIs. A block is fit when its gene has at least 25 gene-assigned UMIs in at least two qualifying pseudobulks. Only ECs whose compatible isoforms map uniquely to that gene enter the local likelihood.

Bounded L-BFGS optimizes the paired multinomial likelihood, with  $\|\delta_{bi}\|_\infty \leq 20$  to prevent floating-point underflow, where  $\|\cdot\|_\infty$  is the maximum absolute component. Two-path fits are initialized from a log-ratio grid because the paired-primer normalizers can make the objective non-concave.

For the cell-resolved variant, sparse cell-by-EC matrices are retained and one local isoform matrix is decoded at a time. The optimizer dimension is still only the number of paths minus one. Cell-specific Fisher contributions are accumulated analytically, so the method does not fit one free splicing parameter per cell.

## Identifiability and reliability

Finite Fisher covariance is not sufficient near a simplex boundary. Parametric bootstrap identified two reliability requirements:

$$\lambda_{\max}(V_{bi}) \leq 1/8 \quad \text{and} \quad \min_s \hat{\psi}_{bis} \geq 0.15.$$

Thus every tested log-ratio direction has effective information of at least eight and every fitted path has at least 15% usage. All estimates are written, together with convergence, identifiability, and reliability flags; only reliable estimates enter GLS. Here  $\lambda_{\max}(V_{bi}) \in \mathbb{R}_+$  is the largest eigenvalue of the  $d_b \times d_b$  covariance, and  $\hat{\psi}_{bis} \in [0, 1]$  is path  $s$ 's fitted proportion.

## Results

### Condition-test scope

The primary analysis used the 25-gene-UMI cutoff and blocks with at most eight represented paths. It fit 10,100 blocks and 359,713 pseudobulk-by-block path estimates; 359,711 fits converged. After reliability filtering, there were 829 conditional and 506 profile tests.

These 1,335 rows represent 829 distinct block-by-cell-type hypotheses, covering 527 distinct splice blocks in eight cell types. The profile analysis is available for a subset of the conditional hypotheses. The possible conditions were 5xFAD, 5xFIRE, 5xFIRE PBS, 5xFIRE transplant, FIRE, and WT. For the 829 conditional hypotheses, 275 compared two conditions, 155 compared three, 153 compared four, 130 compared five, and 116 compared all six. The included set varies because every condition in a test must have at least three independent mouse pseudobulks for that block and cell type.

### Condition-test results

No block was significant at 5% FDR in either covariance analysis. At the unadjusted  $p < 0.05$  level:

- the conditional analysis identified 24 block-by-cell-type hypotheses, covering 21 distinct blocks;
- the profile analysis identified 20 block-by-cell-type hypotheses, covering 17 distinct blocks;
- 20 hypotheses were nominally significant in both analyses.

The union is 24 nominal block-by-cell-type hypotheses covering 21 blocks, but the corrected number of significant hypotheses and blocks is zero. Thus there is no FDR-significant evidence for condition-dependent splicing within cell type. The two covariance variants should not be added as if they were separate discoveries.

## Formal mouse-clustered models

The shared-condition analysis was repeated with the multinomial-logit GLMM and Liang–Zeger GEE defined above. Clustered Gaussian ILR GLS and the unclustered Dirichlet–multinomial were refit to the same canonical partitions as comparators. Each block used 50 synchronized mouse-level null permutations. For 88 two-condition blocks, 50 semi-synthetic replicates were generated at each ILR effect size; 4,369 of 4,400 replicate designs had full rank. “Perm. nominal” below is the number of observed partitions with an event-matched permutation p-value at most 0.05. Table 1 summarizes the complete analysis.

Table 1: Calibration and power comparison. Null rejection uses asymptotic p-values on permuted labels; power uses event-matched permutation ranks.

| Method                | Tests | Conv. | Null < .05 | Perm. nominal | Power .25 | Power .50 |
|-----------------------|-------|-------|------------|---------------|-----------|-----------|
| Clustered GLS         | 359   | 359   | 0.025      | 17            | 0.171     | 0.322     |
| Dirichlet–multinomial | 359   | 359   | 0.138      | 18            | 0.188     | 0.364     |
| Liang–Zeger GEE       | 359   | 355   | 0.415      | 8             | 0.139     | 0.268     |
| Laplace GLMM          | 359   | 359   | 0.108      | 19            | 0.199     | 0.390     |

The canonical GEE sandwich reference is unusable at the available mouse counts: 41.5% of permuted-null p-values were below 0.05, and its apparent 116 asymptotic FDR discoveries are artifacts of this inflation. The GLMM reference was closer but still anti-conservative at 10.8%, while clustered GLS was conservative at 2.5%. Event-matched permutation calibration is therefore required for all comparisons. On that scale, the GLMM had the highest power, improving over clustered GLS by 2.8 and 6.9 percentage points at ILR effects 0.25 and 0.5. GEE was less powerful than clustered GLS after calibration. No method produced a permutation-calibrated 5% FDR discovery in the observed condition analysis.

## Cell-type effects within mouse

The initial Gaussian analysis tested 369 conditional and 218 profile partitions. It found 22 and five FDR-significant partitions, respectively, with all profile discoveries contained in the conditional set. After canonicalizing annotation intervals that encode the same isoform-to-path partition, these 22 intervals represented 14 distinct partitions.

The compositional analysis used converged, conditionally identifiable estimates and required every modeled path proportion to be at least 5%. This produced 646 nonredundant tests. A minimum of 30 pseudobulk observations per partition, chosen as a label-effect-independent coverage requirement, retained 406 hypotheses for FDR correction. Among these, 149 had exact permutation  $p < 0.05$ , and **101 splice-path partitions in 95 genes passed 5% FDR**. Thirteen of the 14 nonredundant Gaussian discoveries were recovered; the remaining partition had exact  $p = 0.032$  but FDR 0.098. Significant compositional results had median maximum cell-type logit effect 0.84 and median effective path count 21.7.

The observation-count threshold is not a narrow boundary effect: thresholds from 29 through 32 pseudobulks each gave 101 discoveries, while 33 gave 99. All 646 observed fits converged. The median number of valid permutations was 500; the minimum was 476 after excluding failed permutation refits. Asymptotic Dirichlet–multinomial p-values were anti-conservative, with 15.5% of permutation-null values below 0.05, which is why only exact per-event permutation ranks are used for the primary result.



Two sensitivity analyses did not improve the event-level result. Testing 3,708 individual cell-type pairs gave 97 FDR-significant pair hypotheses but only 32 distinct partitions after correction across all pairs. Restricting each block to its two molecule-weighted dominant paths increased coverage; with a 1% path floor it gave 88 discoveries, and removing the floor gave 91, under a cross-fitted empirical null. These analyses support the full-path exact omnibus test as the primary cell-type analysis.

Allowing the alternative to re-estimate  $\kappa$  also left the result unchanged: it found the same 101 partitions in the same 95 genes, with no event entering or leaving the 5% FDR set. Exact p-values from fixed- and free-concentration fits had Spearman correlation 0.9995, and both had 148 nominal events. The null optimum was multinomial for 499 of 646 tests, and the alternative optimum was multinomial for 580. Among 66 tests with finite concentration in both models, the median alternative/null concentration ratio was 1.54, consistent with cell-type mean effects explaining some apparent overdispersion. Exact permutation calibration absorbs this nuisance-parameter flexibility, so freeing  $\kappa$  provides no net power gain here.

## Calibration

Across the 25-UMI analysis, 2.7% of permuted-label Wald p-values were below 0.05 for both conditional and profile covariance, indicating conservative null calibration. A parametric bootstrap of 3,000 reliable local fits gave mean squared standardized error 1.10 and 94.5% coverage for nominal 95% intervals. Independent Gaussian simulations of REML-GLS gave type-I error between 2.7% and 5.0% across the tested biological-variance settings.

Numerical tests additionally verify skipped-exon block construction, identifiable covariance after collapsing indistinguishable isoforms, rejection of information-null path contrasts, the analytic binomial Fisher variance, and recovery of imposed GLS effects.

## Coverage-cutoff sensitivity

| UMIs | blocks | tests | conditional null $p < .05$ | profile null $p < .05$ | bootstrap coverage |
|------|--------|-------|----------------------------|------------------------|--------------------|
| 25   | 10100  | 1335  | 0.027                      | 0.027                  | 0.945              |
| 50   | 5896   | 755   | 0.031                      | 0.030                  | 0.936              |
| 100  | 2820   | 283   | 0.029                      | 0.025                  | 0.946              |
| 200  | 1034   | 69    | 0.028                      | 0.026                  | 0.948              |
| 500  | 195    | 10    | 0.030                      | 0.050                  | 0.946              |

No cutoff produced an FDR-significant block. The 500-UMI calibration estimate is based on only 100 permutation tests.

For reliable two-condition, two-path conditional tests, mean asymptotic power for a 0.5 ILR effect was 0.49, 0.51, 0.44, 0.56, and 0.59 at increasing cutoffs. The final two values are selected from only eight and two eligible tests. Among 103 hypotheses shared by the 25- and 50-UMI analyses, power was 0.453 and 0.449. Lower coverage cutoffs therefore increase the number of testable blocks without degrading calibrated precision among shared tests. The 25-UMI cutoff is the default.

## Does cell resolution increase condition-test power?

The cell-resolved likelihood was run on the 527 blocks testable in the 25-UMI primary analysis. All 41,242 block-by-pseudobulk fits converged; 21,624 passed the reliability rules, compared with

21,010 pseudobulk conditional estimates on the same blocks. This produced 836 condition tests, compared with 829 for pseudobulk fitting. No cell-resolved test passed 5% FDR. Condition-label permutations were conservative, with 2.49% of null p-values below 0.05.

An exact matched comparison used the same reliable samples for both methods in 807 block-by-cell-type tests. Cell resolution reduced the median local covariance trace by 1.6%, and the median path-response RMS difference was 0.033 ILR units. P-values were strongly concordant (Spearman rank correlation 0.958); 24 pseudobulk and 25 cell-resolved tests were nominally significant.

For the 259 matched two-condition, two-path tests, mean asymptotic power for a 0.5 ILR effect was 0.504 for both methods. The median ratio of cell-resolved to pseudobulk condition-coefficient standard error was 0.995. Retaining cell likelihood terms therefore did not materially increase power in this dataset. The aggregated EC counts and decoder baseline preserve nearly all relevant local information within a cell type, while residual between-mouse variation dominates the small reduction in inferential variance. Pseudobulk fitting remains the default; the cell-resolved method is available for datasets with stronger within-pseudobulk isoform heterogeneity.

## Compositional and joint-test assessment

Collapsing equivalent isoform partitions reduced the 829 conditional block-by-cell-type rows to 505 nonredundant tests over 317 partitions. The median effective path count was 32 molecules, or 62% of gene UMIs; only 0.6% of estimates reached the gene-UMI cap.

On the same 505 tests, conditional GLS had 12 nominal results and the Dirichlet–multinomial had 18. Neither had a 5% FDR discovery. The Dirichlet–multinomial was conservatively calibrated: 3.76% of 10,100 permuted-label p-values were below 0.05. In contrast, pure multinomial regression gave 197 nominal and 155 FDR results, but 39.1% of its permutation p-values were below 0.05. UMI sampling alone therefore cannot represent the large between-mouse compositional variation. Output tables mark pure multinomial and naive shared-effect rows as invalid for inference while retaining them as calibration diagnostics.

ACAT across cell types produced 8 nominal GLS and 12 nominal Dirichlet–multinomial partitions among 317 tests, with no FDR discoveries. Combining distinct partitions within genes gave 7 and 11 nominal genes among 276 tests, again with no FDR discoveries.

Naive shared-effect pooling illustrates the pseudoreplication risk. It gave 43 nominal and one FDR Dirichlet–multinomial partition, but synchronized mouse permutations rejected 13.2% of nulls at 0.05. The apparent discovery had asymptotic  $p = 6.4 \times 10^{-5}$  but permutation  $p = 0.143$ . Naive shared GLS was also inflated, with 8.4% null rejection.

The mouse-clustered shared GLS restored conservative calibration: 2.19% of 7,169 synchronized permutation p-values were below 0.05. It produced 11 nominal partitions among 359 tests and no FDR discoveries; its smallest p-value was 0.0126 with FDR 0.86. Gene-level ACAT gave 9 nominal genes among 313 and no FDR discoveries. Thus the calibrated gain is limited to the increase from 12 to 18 nominal within-cell-type tests under the Dirichlet–multinomial. None of the evaluated strategies creates a genome-wide condition discovery in this dataset.

## Limitations

The profile covariance is conditional on the fitted genome-wide decoder. Uncertainty in shared decoder parameters is expected to be small for common paths but may matter for rare genes or cell types. A mouse-level jackknife of the hierarchical fit is the preferred extension.

The current biological variance is isotropic in log-ratio space. Estimating a full covariance separately for every block would be unstable with few mice. A later empirical-Bayes model could

shrink path-specific biological covariances across genes.